ORIE 5530
Project: NYC Citi Bike

Fall 2026

Due: December 1st 2026, 11:59pm

Data:

In this project, you will work with NYC Citi Bike trip data for **August 2026**. Download the archive `202608-citibike-tripdata.zip` from

https://s3.amazonaws.com/tripdata/index.html.

Use the NYC archive, not the file whose name begins with `JC-`, which contains Jersey City data. The direct download is `202608-citibike-tripdata.zip`. The archive contains multiple CSV files; combine all of them to obtain the full month's dataset.

Each row corresponds to a ride. The columns include the ride ID, rideable type, start and end times, start and end station names and IDs, locations, and membership status. A description of the published data is available at https://citibikenyc.com/system-data. Python and pandas are strongly recommended, but you may use other programming tools.

First, you will answer warm-up questions to familiarize yourself with the dataset and review probability. You will then use Markov chains to study Citi Bike stations and build a simulation to investigate a question of your choice. The project is open-ended: different groups may develop different defensible approaches.

Warm-up questions:

Clean the data by eliminating rides lasting more than three hours and rides with negative duration, if any. Long rides may correspond to bikes left undocked, and negative durations may reflect recording errors. Report the number of rows and columns before cleaning, the number of rides removed by each rule, and the number retained. Document any additional cleaning decisions.

1. Using the start and end times, compute each ride's duration in minutes and plot a histogram of ride durations.
2. What is the expected ride duration (the average ride duration)? What is the empirical variance of ride duration? What is the probability that a ride lasts more than 20 minutes?
3. What is the probability that a ride lasts more than 20 minutes conditional on the user being a Citi Bike member?
4. Suppose that a ride lasts more than 25 minutes. What is the probability that this ride belongs to a Citi Bike member?
5. What is the expected ride duration of an electric bike? What is the expected ride duration of a classic bike?
6. Suppose that a ride lasts less than 10 minutes. What is the probability that it uses an electric bike? What is the probability that it uses a classic bike? Comment on the results.

Part I: Modeling Citi Bike stations using Markov chains

We would like to study the number of available bikes in a Citi Bike station using Markov chains. For example, suppose there are 20 bikes at a station near Central Park early on a Monday morning. Our goal is to estimate the number of available bikes later that morning and to understand the station's longer-run behavior.

Choose **three popular Citi Bike stations in NYC** with substantial activity. Explain your selection. Split the day into two blocks of time, such as morning and evening, and focus only on weekdays. Specify the hours included in each block. Discretize each block into periods of 5 or 10 minutes, or justify another choice.

Model the number of available bikes at a station as a discrete-time Markov chain. A natural state space consists of the integers from zero to the station's capacity (number of docks). You can find the number of docks in Google Maps. Use arrivals and departures during each time interval to estimate the transitions. Estimate a separate transition model for each station and each time block, using the August weekday data.

This is an open-ended modeling problem. You should make and justify your own modeling choices, state your assumptions clearly, and explain how your model connects to the data. You may propose a different formulation if you explain why it is appropriate and how it addresses the project's objective. In your report:

1. Define your state space, time step, and transition mechanism. Explain how you estimate the Markov Chain from the data and how you handle situations not fully specified by the observations.
2. Present the estimated transition matrices, or a concise representation if they are large. Explain how they differ across stations and time blocks.
3. Compute and interpret the Markov Chain stationary distributions. Discuss whether a stationary distribution exists and is unique for your model; if it does not, explain why and provide an appropriate alternative summary.
4. Discuss what your results imply about station activity or bike availability. Discuss the long-run behavior of each station. Discuss the limitations of these conclusions if any.

Part II: Simulation

Write code to simulate your model from Part I for each station and time block. Choose a starting state and, at each time step, randomly select the next state using your estimated probabilities. Repeat this process many times to study the possible outcomes. State your assumptions and the number of simulation runs.

1. Plot the results of a few simulation runs for each station and time block. Your group should choose a probability or average to estimate using the repeated runs. Clearly define your chosen quantity, explain why it is of interest, and interpret your estimate.
2. Use your simulation to recommend a change that could improve bike availability. For example, would moving five bikes between two stations help? Compare the results with and without your proposed change, quantify the benefit, and explain when it might not help. You may choose another practical question suited to your model, but your answer should lead to a recommendation supported by your results.

Submission requirements:

- This is a group project. Each group should contain at most three students.
- Each student must be linked to a group report on Gradescope. Choose group submission and add your teammates to the same submission. Write all group members' names on the report.
- You should also submit your programming code along with the report (for example, a `.py` file or a notebook file).
- **Report:** Include the warm-up answers, modeling choices and assumptions, results from Part I, and Part II. Present the key results as figures or tables with explanations.
- **Code:** Provide a clearly identified main script or notebook and a short `README` describing the required software, data files, and steps to reproduce the report's figures and tables. Record the settings and random seeds used for the reported results.
- **AI use:** You may use generative AI tools to assist with coding and to improve or streamline the writing of your report. Any such use must be clearly acknowledged in the final report, including a brief description of how the tools were used. The project's approach, reasoning, and conclusions must be developed and understood by the students, and each group is fully responsible for the accuracy and quality of everything it submits. Submitting AI-generated material that the group does not understand, cannot explain, or has not carefully verified will be penalized.
- **Oral explanations:** We may ask any group member to explain any part of their group report orally. Every group member should be prepared to do so, although we will not necessarily request an oral explanation from every student or group.
- **Evaluation:** The project will be evaluated on mathematical correctness, the reasoning behind your modeling choices, the quality of analysis and experiments, reproducibility, and the clarity of writing. Different well-justified approaches are welcome.